

Indian Banking & Financial Requirements Analysis

Domain: Indian Financial Services (RBI, NPCI, UIDAI, SEBI, DPDP) | Model: Local 4-bit Quantized LLM

Parameter	Evaluation Result	Governance Impact
MoSCoW Priority	Must Have	Direct statutory enforcement under RBI/NPCI/UIDAI/DPDP mandates. Failure to implement invites regulatory penalties and license sanctions.
Risk Classification	HIGH (75/100)	Critical data asset; requires dual human & audit approvals
Primary Category	Digital Personal Data Protection & Privacy (DPDP Act 2023)	Subject to cross-border data protection & banking secrecy

1. Original Engineered Requirement

"The system must implement data privacy controls that satisfy GDPR and relevant U.S. financial services regulations, including data minimization, consent management, right to access/erasure, encryption, access controls, and incident response."

2. Regulatory & Compliance Grounding (RAG Citations)

Framework	Compliance Level	Applicable Domain Clauses / Citations
RBI Payment Data Localization Directive (2018)	Advisory Regulatory Grounding	Statutory Citation: RBI/2017-18/153 DPSS.CO.OD No. 2785/06.08.005/2017-18 (Storage of Payment System Data) Matched Terms: Domain Context
RBI Digital Lending Guidelines (2022) & DLG	Advisory Regulatory Grounding	Statutory Citation: RBI/2022-23/111 DOR.CRE.REC.66/21.07.001/2022-23 (Guidelines on Digital Lending) Matched Terms: Domain Context
Digital Personal Data Protection Act, 2023 (DPDP Act)	Mandatory Statutory Enforcement	Statutory Citation: Digital Personal Data Protection Act, 2023 (Act No. 22 of 2023), Sections 6, 8, & 12 Matched Terms: privacy, consent, erasure

3. Technical Controls & Architecture Specification

Control ID	Title & Specification	Automated Verification Method
RBI-DSC-01	Two-Factor Authentication (AFA) & Dynamic Linking Enforce Additional Factor of Authentication (AFA) for all digital payment transactions with dynamic linking of transaction value, beneficiary VPA/account, and time-bound OTP (RBI Master Direction).	Automated security tests verifying OTP invalidation after 180 seconds, max 3 attempts, and zero bypass on dynamic beneficiary binding.

UIDAI-ADV-02	Aadhaar Data Vault (ADV) & Redaction Engine Store 12-digit Aadhaar numbers strictly inside an isolated Aadhaar Data Vault with AES-256 HSM encryption. Mask first 8 digits ('XXXX-XXXX-1234') in all presentation, reporting, and database logs.	CERT-In empaneled auditor inspection, automated regex DB scan for unmasked 12-digit Aadhaar numbers, HSM key rotation audit.
RBI-LOC-03	Payment Data Localization & Sovereign Residency Store all end-to-end payment transaction data, settlement ledgers, and customer logs exclusively on physical infrastructure located within the territory of India (RBI 2018 Directive).	System Audit Report (SAR) by CERT-In empaneled auditor submitted annually to RBI; cloud network egress inspection confirming zero foreign persistent replication.
DPDP-CON-04	Digital Consent Manager & Grievance Redressal Pipeline Implement explicit, revocable multilingual consent collection (available in English and 22 Eighth Schedule languages) with automated statutory grievance resolution tracking within 72 hours.	Consent lifecycle automated integration test verifying immediate downstream API data processing cessation upon consent withdrawal.

4. Traceability Matrix & Recommended SDLC Workflow

Req ID	Target Sub-Module	SDLC Phase / Gate	Verification Deliverable
REQ-IN-FIN-001	NPCI / RBI Payment Gateway Switch	Sprint 1-2 / Cryptographic Architecture	NPCI Conformance Spec & CoFT Token Vault Schema <i>NPCI Sandbox Protocol & Simulator Conformance Certification</i>
REQ-IN-FIN-002	UIDAI Aadhaar Data Vault & V-CIP	Sprint 3 / Identity Security Spike	ADV Token Table Schema & Geotagged Video Store <i>CERT-In Empaneled Security Audit & Liveness Red-Team Test</i>
REQ-IN-FIN-003	FIU-IND & Regulatory Reporting Daemon	Sprint 4 / Regulatory Sprints	FINnet 2.0 XML Schema Exporter <i>FIU-IND Batch Validator Schema Verification</i>